Max Millien

Greater Boston Area · (617) 224-7372 · max.millien@puretome.com · github.com/maxmillienjr

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SUMMARY

Full-stack architect and product owner with experience in turning complex ideas into scalable, production-ready systems. Specialized in RAG architectures, agentic workflows, and high-performance frontend engineering. Expertise across backend infrastructure and UX/UI design, building systems that are technically robust and intuitively usable. Innovating at the intersection of engineering and product strategy: designing for scale, driving engagement, and shipping with speed.

SKILLS

AI & DATA INTELLIGENCE

Generative AI (GenAI) · Hybrid RAG (Neo4j, PGVector) · LangChain/LangGraph · Agentic Workflows & Agentic IDEs (Claude Code) · MCP · Context Engineering · Gemini API · Vertex AI · Ollama · Weaviate

BACKEND & DATABASES

Node.js (NestJS, Express) · C# (.NET Core, ASP.NET) · Java (Spring Boot) · Ruby on Rails · PostgreSQL · MongoDB · Cloud SQL · SQL Server

COMPLIANCE & SECURITY

HIPAA/BAA Compliance · Google DLP (PII De-identification) · Secure WebSockets · JWT/OAuth2 · AES-256 Encryption · Stripe API

INFRASTRUCTURE & EVENTING

GCP · AWS · Azure · Terraform · Kubernetes · Docker · CI/CD · Redis (Bull) · Google Cloud Pub/Sub · QStash · RabbitMQ

FRONTEND & UX ENGINEERING

React · Next.js · Angular · TypeScript/ESNext · Redux Toolkit · Tailwind CSS · shadcn/ui · Spartan UI · Material Design · Slate.js · SEO/GEO · D3.js

OPEN SOURCE

CONTRIBUTIONS · Merged fixes to Angular (core) and claude-mem

PROJECTS · secure-data-vault · agent-native-monorepo · realtime-voice-infra · royalty-lookout

EXPERIENCE

Quiet Horizon PLLC · Principal Architect (C2C contract)

2026 – Present

- Shipped a HIPAA-compliant telehealth and patient dossier management platform utilizing GCP, Google Workspace BAA (ADC), and strict AES-256-GCM encryption for all PHI at rest via NestJS and Drizzle ORM.
- Engineered an ambient AI medical scribe utilizing Vertex AI (Gemini 3). Designed a stateless inference pipeline that transforms raw WebSocket audio dictations into structured psychiatric notes dynamically in-memory, ensuring zero draft-state database writes to preserve strictly immutable medical audit trails.
- Developed a highly reactive clinical workspace using Angular 21 and Spartan UI, featuring a real-time secure dictation pipeline streaming live audio to Google Speech-to-Text, reducing provider charting time by synthesizing complex psychiatric evaluations in seconds.

PureTome · Principal Engineer & Product Owner

2025 – Present

- Spearheaded the product development and UX design to launch an AI Biographer platform, empowering users to craft, publish, and distribute ebook/paperback memoirs and biographies.
- Built and deployed an LLM-agnostic stateless engine leveraging LangGraph, Neo4j, and PGVector, significantly improving large narrative coherence (500k words) over standard RAG implementations.
- Engineered a high-performance monorepo integrating a React 19 frontend with Node.js/NestJS microservices.
- Built an internal growth engine (Next.js, shadcn/ui) to monitor telemetry and orchestrate AI credit economies.
- Designed and shipped a "Hybrid Memory System" combining vector similarity search with graph-based causality mapping to execute multi-step AI workflows without hallucination.

- Orchestrated background job queues and asynchronous service communication using Redis (Bull), Google Cloud Pub/Sub, and QStash for reliable, heavy-workload AI processing.
- Implemented rigorous DevOps and testing standards, building a zero-downtime CI/CD pipeline via GitHub Actions, GCP, and Terraform, backed by comprehensive Playwright E2E and 85% Jest unit test coverage.

NOTABLE PREVIOUS ENGINEERING LEADERSHIP & FULL-STACK ROLES

SBE Vision · Engineering Lead

- Led a team of 11 developers (across 3 time zones), directed agile processes, and architected microservice infrastructure across GKE and on-prem Kubernetes clusters, integrating GitLab CI/Nexus pipelines.
- Designed and developed a modular Angular UI toolkit leveraging Material Design/Tachyons/D3.js, standardizing UX and cutting design-to-development time. The service layer consisted of 10+ microservices built with Spring Boot, Nest.js, and Koa.js, along with Postgres for persistence.
- Wrote an automation framework using Node/MongoDB that used Testcafe and Maven Surefire to provide nightly results on e2e testing of microservices. Service logging was performed using the ELK stack with Filebeat.
- Configured registries for Docker, npm, Helm, and Maven, streamlining DX and cutting build times.
- Implemented an out-of-stack identity provider/management service for authentication (OAuth2, JWT).

Verizon · Senior Software Engineer

- Built web apps and RESTful web services to manage enterprise private cloud organizations and their associated accounts, virtual machines, storage, firewalls, billing, and usage (Dropwizard, Swagger, RabbitMQ, Angular, Express, Nginx, Postgres, CentOS).
- Championed best practices for Node.js, Angular, and Docker adoption, standardizing front-end workflows and improving developer experience.
- Designed cross-platform Vagrant/Docker dev clusters that reduced environment setup time from 2 days to under 2 hours, accelerating team onboarding and CI pipeline readiness.

- Built an automated OpenShift STI deployment pipeline for containerized microservices, enabling zero-downtime releases and reducing deployment errors.

G4S · Senior Software Engineer

- Developed a configurable survey and incident reporting tool (Knockout.js, RavenDB, ASP.NET MVC) deployed across 17K+ security systems, improving data collection efficiency and UX consistency.
- Built SQL-to-RavenDB migration utilities and Excel importers supporting 100K+ system records, enabling seamless legacy-to-modern data transition.
- Enhanced Card Access Control app with 10+ new features and performance optimizations, reducing load times and improving system uptime.
- Developed recruitment workflow module (Angular, C#, RabbitMQ, Azure), implementing CORS architecture, improving request processing speed, and reducing code complexity.

Partners Health · Senior Software Engineer (contract)

- Developed applications for the Acute Care Documentation team to provide real-time electronic medical record solutions, thereby replacing paper flow sheets, assessments, and clinical notes.
- Implemented a cost analysis tool for the Imaging Finance department that is used for tracking the usage of radiology resources owned by Massachusetts General Hospital.

EDUCATION

Boston University

Bachelor of Science, Electrical Engineering